CSE 543: Information Assurance & Security

Detection and Mitigation of Facial Forgeries using Artificial Intelligence in Social Media

Group 6 - Course Project Report

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1. Introduction

The fast progress of artificial intelligence (AI) and deep learning technology has changed the digital media world significantly and even allowed the development of highly realistic synthetic content called deepfakes. However, their malicious use brings severe risks to personal privacy, societal trust, and digital authenticity. In social media, the use of face swaps has been the weapon to convey false information, copy characters, and create materials that are non-consensual or defamatory.

Deepfake detection is based on deep learning models that can spot very small facial inconsistencies and temporal artifacts which are not even seen by human eyes. Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformers have shown great promise in the detection of these manipulations.

Combining blockchain technology and deep learning-based detection is a new security method that has recently become popular. Blockchain gives digital media the property of being unchangeable, meaning that their sources can be permanently verified and their trail of ownership consistently followed. That way, sharing of content can be done with authenticity and accountability.

The project, called Detection and Mitigation of Facial Forgeries using Artificial Intelligence in Social Media, is going to develop and carry out an advanced detection framework new to the market called HODFF-DD.

1.1 Motivation and background

The impersonation of real persons through AI-generated forgeries is now so perfect that it has resulted in:

- Theft of identity and impersonation
- Misleading information dissemination and fake news propagation
- Unauthorized or revenge media
- Harm to one's career and reputation

In addition, the integrity and provenance of digital content are still the issues which, on the other hand, are already identified as major challenges for the stakeholders in the industry. Thus, this endeavor is driven by the demand for a combined AI–Blockchain solution that guarantees:

- Precision and quickness in spotting facial forgeries
- Safe acknowledgment of content's origin and integrity

Pandian et al. [23] and Heidari et al. [35] describe blockchain systems that record a file's history on a shared ledger - once a file is listed as verified, the ledger prevents later changes.

1.2 Goals and Scope

Our project, "Detection and Mitigation of Facial Forgeries using Artificial Intelligence in Social Media," aims to:

- Facial Forgery Detection: Utilize deep learning pipelines that rely on temporal (LSTM/RNN/Transformer), spatial (CNN), and frequency/optical flow cues to detect manipulated faces within photos and videos. [2][10][11][12][18][25][27][29]
- Improve the security of the platform by introducing AI-aided monitoring and real-time screening to flag, throttle, or queue suspicious media. [4][10][13][17]
- Safeguard Data & Privacy: Reducing harm and bias by protecting user data, detecting telemetry through privacy-aware processing, secure storage, and monitoring fairness and authenticity. [1][19]
- Provenance & Identity Verification: Establish dependable blockchain-based techniques for recording cryptographic hashes and detector attestations. [7][8][14][23][26][31][35]

The project investigates dataset bias, and cross-platform degradations in the design, development, and implementation of AI-based deepfake detection and a proof-of-concept blockchain provenance layer. Our team members, under the guidance of **Group Leader Aviroop Nandy** and **Deputy Leader Swapnil Toppo**, are committed to meeting such objectives as agreed upon in the project plan and timeline.

2. Summary of accomplishments in the project

- The project focused on tackling one of today's growing digital challenges - the spread of facial forgeries on social media - by exploring how the intersection of Artificial Intelligence, Deep Learning and Blockchain technologies can be used to detect and prevent deepfake content.
- The team reviewed various research papers distributed as high-level and in-depth analyses where the papers span topics, including deep learning, blockchain-based integrity systems, multimodal fusion methods, and defensive technology over adversarial attacks.
- The primary goal of the project was to analyze practical and effective techniques that strengthen the authenticity and reliability of social media. The team focused on finding how deepfake models work and be applied to protect media users and maintain trust.
- The literature explicitly identifies how modern deepfake synthesis methods such as Generative Adversarial Networks (GANs), Diffusion Models, Autoencoders and Transformer structures have evolved over time. In order to provide solutions, researchers developed various systems, each with its unique strength in detection of patterns. These methods include Convolution Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Vision Transformers (ViTs).
- Among the several key identification architectures in facial forgeries, the team compared ResNeXt-LSTM, XceptionNet, EfficientNet, frequency-based CNNs, and hybrid CNN-Transformer models, among others. Through this comparison, it became clear that hybrid and multimodal architectures outperform traditional single-feature systems, especially when dealing with compressed or poor-quality social media videos.
- A key accomplishment of the project was the successful implementation of the Hybrid Optimized Deep Feature Fusion (HODFF-DD) model. This model combines InceptionResNetV1 and InceptionResNetV2 for spatial feature extraction with a Bidirectional LSTM for analyzing video frame sequences.

- To improve performance, the team used the Spotted Hyena Optimizer (SHO) on Celeb DF v2 dataset, which enhanced the model convergence and stability to identify manipulated facial content.
- The thorough analysis revealed that deep learning frameworks are adept in detecting minor disturbances in facial movements, lighting effects, and lip synchronization changes that usually humans miss to notice. However, the team also recognized key challenges such as high computational costs, bias in the datasets, and the limited knowledge of difficult neural models.
- The project further explored the area of blockchain that could support the content verification as it helps maintain immutable metadata and tracks content provenance. If integrated with AI-based detection systems, blockchain improved transparency and accountability, although problems still exist for scaling techniques and efficient usage of resources.
- The team's review also surveyed promising new directions, such as frequency-domain analysis, multimodal detection, and adversarial defense mechanisms outlined to make deepfake detection models more resilient to new-age forgery techniques.
- There is a need for explainable and trustworthy AI across all fields, which was highlighted by the research to develop models that not only achieve more accuracy but also provide why and how decisions are made. Ensuring transparency can pave the way to design architectures which are self-learning AI models in real-world applications and legal contexts.
- The following are the summarized outcomes of the project:
    1. A comprehensive study of state-of-the-art deepfake detection and blockchain-assisted methods
    2. Implementation and testing of Hybrid Optimized Deep Feature Fusion model
    3. Insights into how AI, Deep Learning, and Blockchain frameworks can work together for multimodal learning to protect the assurance and authenticity of social media content.

3. Accomplishments of each group member

Varshini Akula

- As a member of the team, I contributed extensively to the deep learning investigation and implementation phases, simplifying complex theoretical architectures into a functional prototype inspired by the selected Hybrid Optimized Deep Feature Fusion (HODFF-DD) baseline.
- I reviewed and analyzed various state-of-the-art frameworks focused on deepfake detection through spatial-temporal modeling, including the Multi-Feature Decision Fusion (FG-TEFusionNet) [11], ResNeXt-LSTM blockchain-assited detection [23], AVT2-DWF audio-visual fusion network [48], and CNN-BiLSTM hybrid video classifiers [3, 34].
- From these studies, I derived best practices for face-centric preprocessing (using MTCNN and OpenCV) and spatial-temporal feature fusion, later applied in the HODFF-DD replication. After the integration of InceptionResNetV1 + InceptionResNetV2 encoders with a Bidirectional LSTM classifier, I optimized real-time detection pipeline for temporal consistency and artifact localization in manipulated videos.
- I implemented and tested the detection pipeline in Python (Keras + TensorFLow) using benchmark dataset Celeb DF V2 [61], following the baseline paper's multi-stage evaluation.
- In-depth reference studied: 2, 3, 11, 21, 23, 35, 48

Nandan Jalli

- I have conducted an extensive literature study in this project to have an idea on the potential of deep learning models, along with blockchain technology, to increase the robustness, explainability, and correctness of deepfake detection in social media platforms.

- The advanced level of the research paper analysis that I have done [5], [6], [17], [18], [29], [30], [41], [42] informed me about the various different types of hybrid models to detect deepfakes such as CNN-LSTM-Transformers, Generative Convolutional Vision Transformer (GenConViT), and CSWin Transformer which are created to target both microscopic irregularities within individual frames and identity discrepancies presented in the manipulated images and videos.
- I had a very detailed review of the papers [6], [17], [29], and [42] that helped me to discover the potential of features in the frequency domain and spatial frequency integration (SFIAD). Besides the topics mentioned above, research on defense techniques like 2D Malafide [17] and RoGA [57] was also done.
- My comprehensive study on various AI and blockchain technology-based architectures led me to the conclusion that CNN-Transformer-based hybrid models followed by blockchain verification processes can not only keep up the detection performance but also securely authenticate the content.
- In-depth reference studied: 6, 17, 29, 42, 47, 48, 49

YJ Kim

- My project involved the use of deep learning, blockchain, and artificial intelligence to identify and hinder deepfake content. Such technologies are progressively becoming necessary to uphold digital authenticity and privacy in social media, video-sharing platforms, and secure communication systems.
- A major milestone in my work was the investigation of blockchain-based architectures that not only increase the loyalty but also the openness of deepfake detection networks.
- The references [31] and [43] helped me to get a more profound insight into the ways of using blockchain to assure the reliability of images and videos and, thus, prevent identity theft and tampering.
- Besides, I dug into [44], which talked about the use of AI for detecting fake news and misinformation, and [49], which implemented CNN-based architectures for facial forgery detection. Their combined reference illustrated that the analysis of

the two modalities, visual and textual data, makes the prevention of digital misinformation more effective.

- I deeply comprehended the idea of machine learning methods such as ensemble and convolutional models in identification of facial forgeries. The main point was about the trustworthiness of the model under different quality and lighting conditions (Kumar et al., 2024; Jagam et al., 2025) [32], [49].
- Performance metrics of various systems were evaluated in our study. We noticed that blockchain-verified AI-based classifiers, which are thus supported by the blockchain, have better long-term trust and traceability and, therefore, can establish a more extended period of trust and traceability than standalone detection.
- Helped to form the concepts, linking the discoveries from all the chosen literatures to the suggestion of a single model where deep learning capabilities provide accurate detection while blockchain capabilities provide verifiable provenance and digital trust.
- In-depth reference studied: 7, 8, 9, 20, 25, 32, 44

Aviroop Nandy

- Being the Leader of Group 6 for our course project, I have played an important role in spearheading our research in "Detection and Mitigation of Facial Forgeries using Artificial Intelligence in Social Media."
- I reviewed in detail 8 reference papers on diverse domains such as Convolutional Neural Networks [1], Temporal Deep Learning [25], Machine Learning classifiers [37], Explainable AI [45], Hybrid forensic AI models [60]. I also did a high-level study of 33 papers on various topics: Ethereum Smart Contracts [7], Blockchain for Provenance [8], Generative Adversarial Networks [10], Video Forensic Analysis [12].
- I went through and analyzed various state-of-the-art deepfake detection technologies, including ResNeXt-LSTM blockchain-assisted detection [23], CNN-BiLSTM hybrid video classifiers [34], Inception-ResNet V1 [15],

Inception-ResNet V2 [11], and many more, thus providing the team with comparisons of studies on accuracy, AUC, and cross-data robustness.

- During the implementation for the project, I combined the dual versions of Inception-ResNet models to obtain a stronger model that stresses efficient multi-scale feature extraction and incorporates more residual layers for deeper normalization. Then, I included a feature-fusion extraction mechanism for the output fusion from multiple feature extraction models.
- After establishing the model, I helped in training the model to test its accuracy. We had chosen the CelebDF (v2) dataset for training, containing over 6000 video samples-genuine and fake, respectively. I performed model loading, data loading, inference execution, and performance metrics calculation.
- In addition, I, as Leader of Group 6, helped in coordinating and collaborating with all the team members to ensure timely submission of project proposals, weekly reports, gantt chart updations, and group meetings.
- In-depth reference studied: 1, 13, 25, 37, 41, 45, 49, 60

Priyanshu Patra

- My work, documented in this paper, constitutes a major step forward in the area of digital media security and deepfake detection. After thoroughly reviewing a wide range of research articles, I have pinpointed, as well as merged, numerous groundbreaking technologies and algorithms in my proposed framework to promote information assurance and media integrity.
- I have implemented and modified Artificial Intelligence (AI) algorithms of high calibers, such as the Swin-Transformer architectures for telling AI-generated images apart [28], the deep Convolutional Neural Networks (CNNs) for detecting forgery in the forensic field with more accuracy [33], and the Facial Action Unit (AU) for recognizing the minute physiologically inconsistent aspects of deepfake videos [52].
- The research network is keen on the point that the energy of the deep learning specialized models can be tapped by building different networks like the FSDNet

for decoupled analysis of the style of the forgery [40] and the deep-learning architectures comparison system [51].

- I have further conducted the comparative analysis to scrutinize the state-of-the-art technologies, methods, and trends concerning multimedia deepfake detection, which acts as an additional layer of scholarly rigor and present market orientation to my framework [10].
- In-depth reference studied: 10, 22, 28, 34, 39, 40, 45

Swapnil Toppo

- As the Deputy Leader of Group 6, I worked closely with the Group Leader (Aviroop Nandy) to coordinate and maintain every stage of the project, ensuring smooth collaboration, timely progress, and integration across all modules.
- I conducted a focused review of research papers on hybrid spatiotemporal deepfake detection, including CNN-LSTM-Transformer models [6], multi-feature decision fusion frameworks [11], and optical-flow-based CNNs [12].
- To be able to identify minute irregularities in face forgeries and capture motion dependencies across frames, I created and built a BiLSTM-based temporal analysis module.
- To increase prediction stability and cross-dataset performance, I created a majority-voting ensemble model that combines BiLSTM and Inception-ResNet outputs.
- I identified dependencies, kept track of milestones for all technical and documentation work, and updated the Gantt chart on a regular basis.
- In order to improve collaboration, communication, and morale during the research and implementation stages, I assisted in setting up and running team meetings.
- In-depth reference studied: 4, 16, 18, 27, 40, 42, 56, 57

4. Detailed Results

4.1 Overview of Information Assurance in Social Media

Yet digital assets are the new form through which people socialize, express opinions, and share their lives, but in a digital world where artificial intelligence runs rampant, it's easier than ever to create false narratives through images, videos, and sounds. Thus, Deepfakes - highly realistic images/videos mediated through AI - pose one of the greatest concerns against truth and credibility in today's digital era and social media like. They can and have perpetuated misinformation and reputational attacks, and it's all too easy for users to understand what's legitimate and what's not. Thus, scholars have turned to deep learning and blockchain technology in an era of social media to authenticate the validity of such information.

Deep learning tends to assess subtle visual and behavioral anomalies that reveal falsified data, while blockchain technology tends to establish a secure, transparent, and ultimately irreversible approach to media attribution and its metadata from the moment of creation. For example, Mastoi et al. (2025) developed a blockchain-based watermarking solution that essentially decentralizes authenticity-based verification across media with reputable origins and not compromised integrity [7]. Colaco et al. (2025) apply blockchain as a detection resource through convolutional and recurrent deep learning architectures to assess video manipulations across social networks and dissemination efforts [8]. On detection, Sadhya and Qi (2023) note a deep learning model based on an attention mechanism to pinpoint unnatural inaccuracies in facial recognition for increased detection accuracy, given less effective compression/editing tools [20]. Finally, Malik et al. (2022) employed a systematic review of various studies examining deepfake detection via deep learning and claims that a more permanent answer by way of artificial intelligence combined with a verification effort is necessary for credibility [19].

Thus, these sources conclude that while no one definitive solution exists to contain deepfakes once and for all, the intersectional solutions between the transparency and immutability of blockchain and the extensive analysis through deep learning provide a

more trustworthy solution of expected assurance. Deep learning and blockchain are crucial for establishing authenticated, verifiable information networks on social media.

4.1.1 Analyzing the Current State of Information Security

The current state of information security related to social media is a bit of a tug-of-war between AI development and discernible reality; with deepfakes increasingly accessible it's difficult to govern what's human-made and what's accessible on social media without overstepping (or understepping) boundaries. Malik et al. (2022) report that the technology behind deepfakes can generate faces and voices so that misinformation, identity fraud and defamation can occur [19]. They report that trends in digital deepfakes typically fall victim to detection as methods of generation outpace currently established models.

In light of this, much research has been done to determine new deep learning models that assess visual and social inconsistencies. Sadhya and Qi (2023) proposed an attention-based model that detects nuanced patterns in facial recognition and eye movement, which systems are meant to capture to appreciate false images/videos [20]. They claim significant performance but note the systems' vulnerability when models are given compressed settings, low-quality images or images that never appear in their datasets.

To promote information assurance, Mastoi et al. (2025) promote a blockchain-based system that assesses media credibility over decentralized ledgers [7]. This champions less detection but more validation so that should a fake image/video be generated, it cannot be rendered as real by any means. In conjunction with Mastoi et al., Colaco et al. (2025) utilize blockchain in a deep-learning model to champion a nexus of detection and credibility that cannot be manipulated once it's in the ledger [8].

Therefore, the current state of information security related to social media benefits from findings but also has easily recognizable limitations. Deep learning systems grow in detection accuracy over time; blockchain generates an indisputable means of validation. Yet these are not widely met in real-world application yet as confidence struggles

against scalability and data quality; not to mention there is no industry standard yet for all media purposes. As deepfakes become stronger, only resilience and durability emerge from how well these systems can adapt over time.

4.1.2 Identifying Vulnerabilities in Social Media Platforms

The greatest vulnerabilities that social media currently faces from this technology relate to human deception and manipulated content that becomes increasingly reliable with AI creations. Malik et al. (2022) note that with the rapid generation capabilities associated with AI that average human users can no longer tell what's real and what's not; thus misinformation proliferates through social media [19]. Sadhya and Qi (2023) note that even systems that can detect falsities cannot engage if videos/images are compressed or generated through synthesis techniques that have yet to be explored [20].

Without means for validation, identity theft and fabricated content generation can occur. Costales et al. (2023) note that blockchain-secured verification levels credible verification with a cryptographic hash at creation's inception so everything ever created is always verifiable as original [31]. Patil et al. (2021) established a blockchain framework to secure verification findings from tampering so that ownership remains intact [43].

Text provides another level of opportunity for deception. According to Ashish et al. (2024), it's significantly more believable to have fake news paired with fake images, creating cognitive dissonance within the general public [44]. Therefore, it's clear that vulnerabilities stem from technological gaps as well as trust gaps from human users that need interdisciplinary efforts across all content-types created.

4.1.3 Strategies for Enhancing Information Assurance

The best way for information assurance to improve is to combine deep learning detection with blockchain validation within a decentralized, collaborative world. Mastoi et al. (2025) argue for a blockchain-supported watermark system where encryption labels become stabilized on a decentralized ledger to support traceable credibility [7]. Colaco et al. (2025) took this assessment further by conducting an experimental study to

combine deep learning detection with blockchain validation for immutably traceable efforts [8].

While assurance from blockchain is best for something never seen before or determined reputationally valid, in the case of recognition against falsification, deep learning rules. Sadhya and Qi (2023) used attentional networks for simple consistency detection; Jagam et al. (2025) employed convolutional neural networks for precision based on resourceful image collections [20], [49]. In real-time, Verma et al. (2025) note blockchain as a trust-promoting video-enabled communication system for transparency against tampering [50].

Finally, the ability to promote digital authenticity comes from a reliable distinction from AI's recognizable assessment to an unmanageable path within blockchain's development. Credibility can be restored to media when combined through interdisciplinary trust on social networks that champion authentic use across platforms.

4.2 Leveraging AI/Deep Learning for deepfake detection

The distinguishing method of characterizing the vast majority of deepfake detection solutions is through AI [5]. As malicious models creating deepfakes become more advanced, so does the need for means of combating and exposing them. In fact, much literature positions AI and deep learning as detection sources [46]. This section discusses what AI and deep learning models are commonly used for Deepfake Image Forensics [1] or Video Authenticity Detection [5].

4.2.1 Basic AI Architectures for Deepfake Detection

The most rudimentary type of architecture in deep fake detection relies on either a CNN or an RNN.

Convolutional Neural Networks (CNNs) for spatial analysis:

The most common AI model to create detection is a CNN trained to acknowledge spatial or visual artifacts in the image that the human eye cannot see, physically [49].

Countless papers utilize Deep Convolutional Neural Network [34] or Convolution Neural Network [33] which achieves effective detection of fake face images and/or forgeries.

Recurrent Neural Networks (RNNs) for temporal analysis:

Since deepfakes most often occur as videos projected in frames over time, some means of detection considers the temporal function of frames. In the case of video, Recurrent Neural Networks or RNNs are successful in processing sequences and detecting unnatural patterns of time [25]. For example, deepfakes may create unnatural blinking movements or inconsistent movements that otherwise wouldn't be detected if forced to analyze a static frame.

Specific CNNs:

There are specialized forms of CNNs that attempt to analyze particular artifacts. A form of detection is Optical Flow Based CNN which observes the flow of motion vectors in sequences and direction that occurs over time that's oppositional in actual vs deepfake video [12]. One common method is Error Level Analysis or ELA which trains a CNN to see differences in compression levels magnitude of images; compression differences occur when digital manipulation does not create the same compression during merged images [60].

4.2.2 Advanced Architectures: Transformers and Hybrid Models

In order to deter more realistic fakes, researchers have sought after more advanced and intelligent models like Transformers and Hybrid Models.

Transformer Models:

When images are AI-generated, Transformers have been found effective in identifying this occurrence [28]. The efficacy of Vision Transformer models, more advanced than other forms, like Generative Convolutional Vision Transformer or GenConViT proves accurate in detection [29].

Hybrid Models:

When combined, the most accurate detection occurs from those of others. The following models are common:

- Hybrid CNN-LSTM-Transformer model that uses CNNs to detect spatial attributes, LSTM to assess processed temporal data and Transformers to understand context [6].
- Hybrid CNN-Transformer Approach where Frequency Analysis aids in detection of artifacts in another domain [27].
- An efficient system combines CSWin Transformer with CNN model for improved results [30].

4.2.3 AI-Driven Feature and Artifact Analysis

The latest studies for detection with AI also target what needs to be detected. Instead of generalized frameworks for detection, many models are created to find specific features or artifacts.

Physiological and Behavioral Cues:

Certain AIs are designed to recognize minute discrepancies originating from physiological indicators. These include real-time deepfake video detection using eye movement analysis [13] and Facial Action Unit-Based Deepfake Video Detection [52].

Frequency Domain Analysis:

The best means by which this occurs is an AI trained network that seeks to analyze frequency domain opposed to pixel level. Frequency CNN [36] or Spatial-Frequency Feature Integration [18] accomplish this.

Multi-Modal and Feature Fusion:

Since deepfakes can also include audio, audio-visual fusion [48] and Hierarchical Contextual Aggregations prove highly effective for detection purposes [58]. The same can be said for multi-feature decision fusion [11] and Boosting Deep Feature Fusion [15], where fusions create more effective deepfake video detection methods [4].

Explainable AI (XAI):

As models grow more complicated, they need to be explainable in terms of how they recognize a video as such. XAI (Explainable AI) needs rigorous deep learning in Explainable DeepFake Detection (XAI) as it's important that clarity about how it works exists [4, 54].

Novel Model Approaches:

Finally, newer literature pushes the envelope with novel architecture model design approaches whether it be 3D EfficientB7 with Spatial Attention [16] or Zero-Shot Visual Deepfake Detection where there's an intention placed on forgery detection prior to the AI ever encountering it [42].

4.3 Leveraging Blockchain Technology for deepfake

Deepfake technology has distorted the very definitions of digital authenticity and, instead of using AI to replicate human faces or voices in private gatherings, artificial videos are sent across social and communicative networks. When this is the case, separating legitimate media from created works is increasingly complicated. While deep learning detection systems may exist, they are not foolproof as an artificially generated image could still pass even the best experts if it is newly generated or highly compressed.

Recently, however, blockchain has become a secondary solution. Its decentralized and impervious properties make it a great potential option for digital media attribution and tracing. Mastoi et al. (2025) declare that with blockchain storage and transparency, future forgery can be avoided - once verification is done and established in the blockchain, a file is unlikely to be forged at a later point [7]. Colaco et al. (2025) note that as social platforms provide audit trails of what the deepfake algorithm finds and assumes, blockchain can act as a secondary audit trail [8].

Ultimately, blockchain and deep learning present a two-tiered approach that champions the falsified materials' location and lends verifiable proof that whatever is authentic, is,

indeed, authentic, regardless of the operating platform. It's no longer just about detection, though; it's about fostering trust.

4.3.1 Role of Blockchain in Ensuring Media Integrity

Blockchain is one means of information assurance through untraceable integrity. Mastoi et al. (2025) suggest a conceptual framework where blockchain watermarking across videos applies encrypted identifiers to the video, allowing in-the-moment ownership identification and existing media rights [7]. Essentially, any change to the file would be detected immediately.

Colaco et al. (2025) take it to the next level in a practical system where blockchain communicates with deep learning classifiers [8]. For instance, convolutional neural networks determine whether the aesthetic fabric of a deepfake video possesses any incongruence before hooking up to recurrent neural networks for validation or denial for more precise detection. If validated, then the written results are transmitted to the blockchain - and nothing, at that point, would ever improperly mishandle the verification trail.

Costales et al. (2023) even further establish the Deepfake Analyzer framework where blockchain allows for videos/images to be confirmed safe against identity theft by smart contracts [31]. Whenever a file is verified, a hash is generated through the distributed ledger of the network; in other words, if someone merely saves an image, then they cannot, later on, have someone else verify it - unless someone else substantially altered it, of course. This means blockchain is not just an outside resource, but cumulative interconnectedness that champions AI detection through cryptographic verifiability.

4.3.2 Blockchain-Based Frameworks for Deepfake Detection

Many systems arise about how best to combine verification and detection to ensure the content is correct in the first place. Patil et al. (2021) explore a decentralized, blockchain-based solution which involuntarily edits videos instead of allowing access for manipulative freedoms [43]. This means that the videos themselves are hosted on blockchains as fingerprints confirm when decentralized memoirs are created. This

means, for incorrect access and evolution, hashes will confuse relative dimensions to those percentages within its time stamped memoir rendering.

Verma et al. (2025) utilize real-time protection for video conferencing systems and accumulate hashes over time on a blockchain to establish ledgerically transparent memoirs of what was sent in each frame [50]. This acts as evidence against impersonation and replay attacks.

Colaco et al. (2025) and Mastoi et al. (2025) hope beyond that the blockchain can act as more than a storage device, but instead actuate communication worthy of details proximate to gains or revelations [7], [8]. For example, if the detection model proves something digitally intelligent, its metadata and effective results go to the chain for forever accessibility; if it proves ineffective later, hashing will recognize a discrepancy.

Ultimately, these frameworks signal how true digital trust is possible through AI strengths and blockchain transparency sustainably when deep learning capability is combined with immutability.

4.3.3 Challenges and Future Directions

Yet despite promising theoretical successes courted by recent efforts relating blockchain solutions to deepfakes, Colaco et al. (2025) understand that integration is complicated, preventing massive awareness across social platforms [8]. For example, indiscriminately maintaining authenticity through a distributed ledger associated with a public blockchain isn't manageable; sanctioned agents offer transactions made on public blockchains, but heavy transaction fees and time-sensitive actions are inefficient relative to social dynamics that boast millions of uploads daily.

Both a lightened blockchain system and attractive privacy measures can help. Patil et al. (2021) suggest less energy-consuming levels of consensus optimal for multimedia verification [43]. Verma et al. (2025) recommend hybridized potentials across public-private blocks - with personal identification data kept private on private blockchains while general access for greater verification via public chains fosters

universal verification [50]. This mixed-reality champions confidentiality while still holding people accountable at preferred access points for safe networking.

Finally, suggest a potential integration with distributed verification possibilities where authenticated links to identities go beyond social platforms through explainable AI layers that provide permanent digital connections of what was once prompted years later for inquiry. Once sufficient digital identities are recognized as accessible, techniques that have developed expertise on how to glean crypto and digital wallets will inevitably be implemented for de-fact verification champions for any image or video that could have/should have been presumed unverified along the way through independent means when social users questioned why they transmitted illegitimacy or illegality.

These new technologies will champion generations of truths while decent creators throughout digital footprints will create second-chance fosters of trust for what previously had been nothing but digital deception for nefarious means.

4.4 Data security and privacy

With deepfakes comes concern over data integrity and security [1]. Methods of forgery have moved from misinformation to personal invasion with the goal of Nicking Identity Theft [31] and further undermining Digital Trust accorded visual data. Thus, the greatest concern here is maintaining the integrity of Evidentiary Integrity [24] for visual data

4.4.1 The Threat to Media Integrity and Privacy

The biggest risk of deepfakes relates to the effective creation of seemingly real, yet false, images of people to attempt defamation and fraud. This violates Privacy Protection [1]. The same concern emerges for critical systems like Facial Payment Systems, for if a deepfake attack works it is seen to have quick Financial Impacts [39]. This emerges because of diminishing Digital Trust [31] and Protecting Image and Video Integrity [31].

4.4.2 Blockchain as a Decentralized Defense

Data integrity with deepfake suggests Blockchain Technology [7, 8] as a mediating

response. Blockchain is a decentralized, non-alterable form of record keeping that could create a chain of custody for digital media.

Integrity Verification:

This technology is used for Blockchain-Assisted Video Integrity Verification [23]. Combating Deepfake Videos Using Blockchain and Smart Contracts [59] shows that with every video creation, the hash is timestamped in the blockchain to allow for some entity to check if a video has been altered [53].

Decentralized Trust:

This information comes from Blockchain-Driven Defense Against Deepfakes on Social Media Platforms [14]. The trust model is decentralized as opposed to centralized trust established. The intention of the Blockchain Technique is to Resist Against Deepfake Videos [43] through a non-changeable, transparent history of creation.

Hybrid Solutions:

Some suggest this should be a dual response with AI-detection generated solutions. This includes Enhancing Deepfake Content Detection Through Blockchain [7] as well as detection models made within Blockchain Environment [26]. These are two levels of security for AI models developed to detect forgery while the blockchain runs Blockchain-Based [35] solutions to validate authenticity.

Broader Implications:

This security solution applies more than social media applications. It applies to Enhancing Security and Transparency in Video Conferencing Systems through Blockchain Integration [50] as it maintains another important domain that could be compromised through live action deepfakes.

4.5 Case Studies and Practical Implementation

The following case studies and implementations position the practical implementations of AI and blockchain as more relatable where perceptions of relative improvements to information assurance (integrity, availability, non-repudiation, privacy) for such social media options manifest clearer.

4.5.1 Implementation of AI-Based Deepfake Detection

To implement practically, a **social media platform** would integrate an **AI based deepfake detection** mechanism.

System design (end-to-end)

- **Ingestion & pre-processing:** Face detection/alignment and tracking occur, video/image sampling into frames occur; face-oriented crops maintain identifiable micro-artifacts while temporal signals are extracted as **optical flow** to display movement inconsistencies effectively from realistic constraints. [12].
- **Modeling:** The resultant **spatiotemporal stack** for frame and clip level classification combines CNN feature extractors with a sequence model (BiLSTM/RNN/Transformers); frequency-domain features (e.g., FFT bands) and feature-fusion increase resilience against compression/unseen generators. [25][27][11][36][18][29][30].
- **Real-time triage:** A lightweight "first-pass" model can be run for near real-time screening to flag cases for more resource-intensive verifiers; eye movement/blinking patterns can serve as stronger features facilitating rapid decisions upon live streaming in addition to facial features. [4][13][16].
- **Hardening:** Based on previous work that focused on adversarial attacks on face deepfake detectors, the team assesses **adversarial attacks** and anti-forensic perturbations so training is reinforced to deal with them and much later in training. [17]. Based on survey-driven suggestions for generalizability cross-dataset assessment and stress testing through codecs and bitrates is performed. [10][19].

- **Actions & UX:** Depending on confidence/classifier policy, the platform adds warnings, can limit distribution, or redirect human review; riskier accounts may need additional vetting. Detectors act inline as detected in **facial payment** evaluations which are critical to payment/identity-sensitive actions. [39].

Why this is important

Frequency cues reveal GAN artifacts and re-encoding didn't mask other findings [36][18]. RNN/Transformer heads provide long-range understanding. [25][27]. **Multi-feature decision fusion** stabilizes predictions across degradations of assessment. [11]. Optical-flow CNN channel reveals temporal inaccuracies. [12]. Ultimately, these findings, aggregated, relate to the most current surveys/literature found regarding the detection systems themselves with an operationally based focus. [2][10][19].

4.5.2 Blockchain-Enabled Media Provenance in Practice

To strengthen **non-repudiation** and **chain-of-custody**, the platform adds a **provenance layer**:

Workflow.

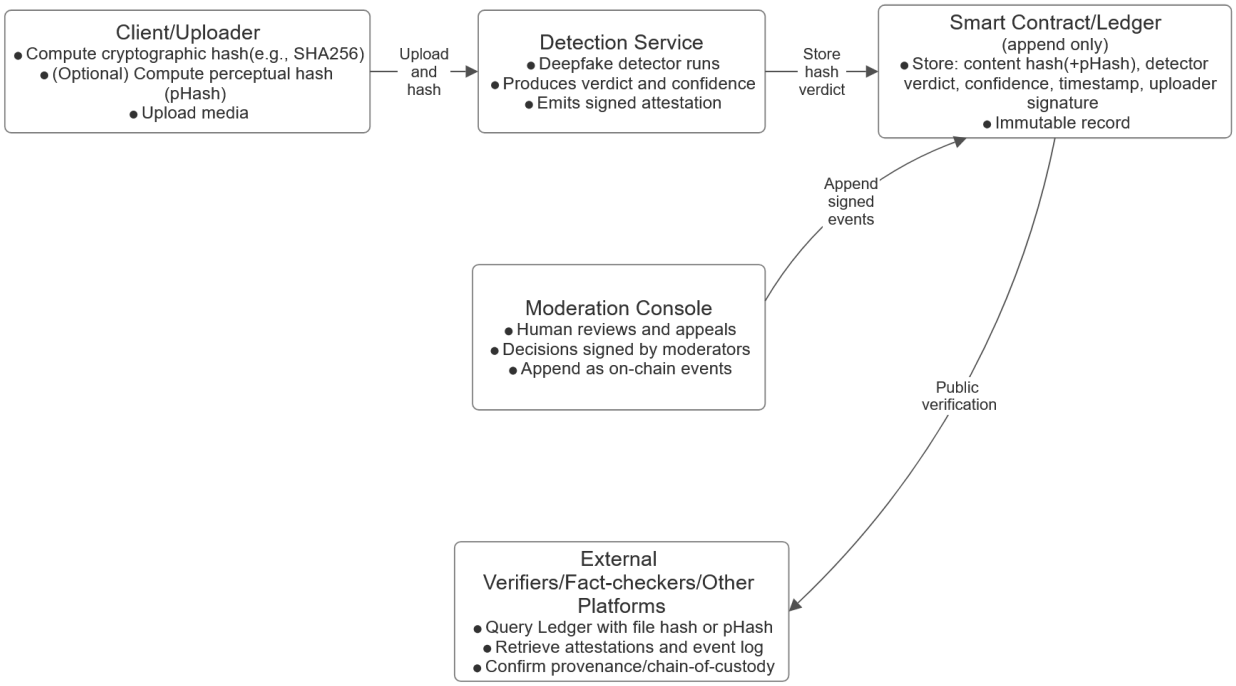

Figure 1: Workflow of Blockchain-enabled Media Provenance

- On upload, the client computes a cryptographic hash of the media; the detection service stores the hash plus model verdict and confidence in a smart contract.
- Moderation updates, such as decisions made by human reviewers and the results of appeals, are included as signed events.
- External verifiers, fact-checkers, and even other platforms can **query the ledger** to confirm whether a given file (or robust perceptual hash) matches a previously verified object.

Applied designs from federated/consortium deployment contexts to in-loop detection ledger and watermarking plus ledger solutions show how blockchain incentivizes detection by **anchoring provenance claims** and detector returns to a distributed and immutable ledger. They tackle cross platform accountability, audit trails and tamper evident characteristics that inherently increase information assurance through social ecosystems.

Development decisions made at the prototype level (i.e., permissioned chain with reduced latency/fees vs. public chain with increased transparency; 205 hashes and attestations uploaded to prevent raw media visibility for privacy reasons establish the **hashes and attestations** that link to avoid access costs between nodes and payloads.

4.5.3 Measuring the Impact on Information Assurance

A comprehensive assessment allows for in-field advantages to be acknowledged:

Detection quality & robustness.

- **Chief KPIs:** AUC-ROC/PR-AUC, F1, EER, TPR at low FPR (e.g., 1%), inference latency, and throughput (clips/sec).
- Robustness panels: detection performance over **novel generators**, extreme compression, resizing, and social-platform features; detection by generalization against **adversarial attacks** and mere post-processing (blur, noise, sharpen) [10][17][19].

Integrity & provenance results.

- **Ledger indicators:** proof-commit latency, verification duration, and coverage (percent of uploads where hashes were recorded).
- Assurance metrics: decreased recurrence of abuse and successful impersonation rates, increased reviewer agreement (detector + human), and provable **chain-of-custody** for long-form references [7][14][23][31][35].

Privacy & fairness protections.

- Assess data-minimization extent (no identifiable faces in archives), bias assessments within demographic subdivisions and creators false-positive implications all relative to recent survey and digital forensics literature privacy/authenticity standards [1][10][19].

Prototypes exhibit in the literature how blockchain anchoring provides quasi-permanent, verifiable provenance whilst **spatiotemporal + temporal + fusion** models extend

robustness against various standard degradations. Together they enhance the **reliability** and **integrity** of social user generated content on social media sites.

4.6 Performance evaluation and key metrics

4.6.1 Assessing System Performance

Deepfake detection and mitigation systems are evaluated based on the extent to which a model identifies real visual media and forgery as such in an open world setting. Yet, most experimental performance assessment in the literature operates through quantitative experiments on open benchmark datasets: **FaceForensics++**, **Celeb-DF** and the **DeepFake Detection Challenge (DFDC)** dataset. Model verifiability is validated through cross-dataset generalization efforts that assess whether models can use what they've learned on additional forgeries from other source domains.

The assessments are generally **frame-based**, **video-based**, or **spatiotemporal-based**. Frame-based characteristics evaluate classification within each frame independently. Video-based characteristics evaluate a prediction made from multiple frames to provide temporal coherence. For example, Zhou et al. [6] found that CNN-LSTM-Transformer hybrid pipelines outperform one-stage detectors due to spatial and temporal dependencies. Similarly, Rajesh and Maruthuperumal [16] found that 3D EfficientNetB7 with spatial attention components provides better results which suggests localized features add to predictive power. Metrics for key performance evaluation include **classification accuracy**, **precision**, **recall** and **F1 score**. Such support from **ROC-AUC curves** respectively illustrate sensitivities and specificities.

In addition to detection accuracy, there are factors influencing the performance evaluation of a system relative to its **efficacy** and **robustness**. For example, Abdul Qadir et al. [4] found that deep learning architecture improvements support enhanced detection over time with stable findings whereas Hasan [5] found that inference in real-time is critical for social media settings. Quantitative metrics that represent **training time, number of parameters and inference time per frame** are all relative to

scalability and deployability efforts, especially when systems must be lightweight for on-device applications.

4.6.2 Key Metrics for Information Assurance

Information assurance refers to the reliability, trustworthiness and stability of AI model rendered decisions as such. Most characteristics are evaluated based on the four pillars of information assurance: **accuracy, robustness, explainability, security**. The following quantitative metrics are commonly reported in the literature:

- **Accuracy (ACC)**: The percentage of true positives and true negatives (real + fake) that are correct.
- **Precision (P)**: The number of fakes detected that are ultimately confirmed as fakes.
- **Recall (R)**: The ability to identify all fakes found in a database.
- **F1 score**: The assessment of precision + recall that provides a balance scoring system for detection quality.
- **AUC**: The area under the curve, relative to true positives vs false positives.
- **Equal Error Rate (EER)**: The location where the false accept rate is equal to the false reject rate; more commonly applied to biometrics + identification systems [19].

Khan et al. [10] suggest generalization metrics which examine whether systems overfit to specific types of fakes since the cross-dataset results must prove otherwise. Lei et al. [11] introduced a decision score based on multi-feature fusion which compiles many dimensional approaches into one assured space. Furthermore, multiple robustness metrics due to *adversarial attacks* - i.e., Galdi et al. [17] assess how a model maintains integrity with accuracy in the face of challenges posed by forensics perturbations.

Information assurance involves how deepfake explainability + trust metrics are rendered visually through Grad-CAM, feature activation mapping and therefore, interpretability for transparency which is where forensic - legal trustworthiness comes into play when detection must be validated for evidentiary trust.

4.6.3 In-depth Analysis for AI Technologies

AI-based deception detection systems are based on statistical deep networks that find spatial, temporal and frequency domain features associated with face-based forensics. Where implications differ based on relative architecture is as follows:

- CNN-based detection InceptionResNet + XceptionNet championed through texture analysis + pixel differences [36].
- RNN/LSTM connections detect movement from frame to frame with improper connections in between or non-human expected movement patterns [25].
- Transformers connect contextually across the entire video and thus, have better generalization + interpretability [6].
- Multi-feature fusion systems combined RGB + frequency + optical flow to be robust against compressive artifacts or low quality. [11].

Statistics about detection success corroborated through empirical findings include Zhou et al. [6] who found a mixed reality AUC of 0.984 on Celeb-DF data attributes with CNN-LSTM-Transformer hybrid; Amerini et al. [12] found CNN optical flow detection at 93% accuracy; Lei et al. [11] with multi-feature fusion at 96.7% accuracy; Malik et al. [19] at the largest generalizability across other trained datasets based on temporal + frequency.

Other metrics worth assessing for performance include:

- Frame Consistency Score (FCS): temporal coherence for prediction across frames.
- Adversarial Robustness Index (ARI): stability based on noise/counter forensics.
- Computation Efficiency (CE): balance of inference speed vs model development.
- Cross-Dataset Accuracy Drop (CDAD): domain generalization potential that remains accurate enough to prevail when dealing with previously unseen data.

The literature champions over statistical detection as model robustness, interpretability and dataset generalization is just as important as ultimate accuracy when defining performance evaluation in forgery detection systems based on AI's quality assessment.

4.6.4 In-depth Analysis for Blockchain Technologies

Blockchain solutions aimed at media integrity reach one goal: authenticity detection. This correlates to source integrity + verification through relative superiority of technological solutions under **performance assessment** for system efficiency/speed and security/assurance/quality assessment for hashing integrity/tamper resistance/etc..

Key solutions supported by research from Qurat-ul-Ain Mastoi et al. [7], Elgalaly et al. [14], Heidari et al.[35] show how blockchain can secure media integrity, authenticate source integrity, decentralized integrity/verification provide confidence/compliance/multi-agent negotiations.

Relative performance metrics include:

Throughput (Transactions per second): amount of records verified and validated over a time span.

Block generation time (average): time until block confirmation.

Gas Consumption: smart contract execution costs from Ethereum solutions [8].

Storage Overhead: data taken up through verified hashes + metadata [31].

Similarly, security-related assessments include:

- Integrity Verification Rate (IVR): amount of integrity verified via hashes.
- Tamper Detection Rate (TDR): unauthorized detection based on criminal activity.
- Consensus Success Rate (CSR): amount of nodes in agreement.
- Resilience Index: stability against attacked situations/selfish node circumvention [35].

Where Mastoi et al.[7], Elgalaly et al.[14], Heidari et al.[35] cite successful integrity validation above 95% with low latency, Heidari et al.[35] support how federated blockchain integrity access improves feasibility since integrity verification isn't relegated to a single source.

Performance assessments in blockchain solutions favor security/traceability/computational efficiency while AI systems celebrate practical application/accuracy/comparative generalization otherwise rarely assessed.

These assessments provide supplemental data fields for comparative systems working together yet often treated independently and provide additional insight for theoretical performance when it comes to making trusted generated media.

4.7 Implication of research findings

The deepfake detection arena is ever-changing, essentially an arms race between forgers and those seeking detection [46]. The technology has implications, challenges, and a future, all of which are noted in a Systematic Review [46] and various Survey [10, 19] and Review [2, 45] articles.

4.7.1 The Evolving Challenge and Current Limitations

The foremost implication of current research is that Generative Artificial Intelligence is An Evolving Challenge [46]. The advances in state of the art generative models can increase accuracy, meaning detection models need to improve, yet there are challenges and limitations [10] that detection models face.

Poor Generalization:

The first and foremost problem with AI detectors is they're Poor Generalizers. They perform poorly on unseen types of deepfakes that are not presented during training [10].

Limited Data:

A comprehensive review [45] notes that Limited Data [45] may impact detection algorithms as these comprehensive deep-learning models rely on expansive data sets.

Adversarial Attacks:

Detection models are susceptible to Adversarial Attacks [10] whereby the users generating the deepfakes create subtle modifications on purpose to generate a

deepfake that gets past a specific AI detector.

Adaptability:

The field is changing all the time, and models must Adapt [45]. A model that works today, may be outdated in three months.

4.7.2 Future Directions and Emerging Trends

Thus, from these challenges, a survey on multimedia-enabled deepFake detection [10] and other reviews note trends [10] and future directions [10] that must be pursued in the field of deep learning.

Focus on Generalization:

A detector should be able to detect forgery without the forger knowing how it's done. This includes Zero Shot Learning [42] and making models resistant to new techniques.

Explainable AI (XAI):

At this moment, whatever we detect becomes more complex and harder to explain why that video was detected in the first place. Current research promotes Explainable Deepfake Detection [54] so that the process can be understood and credible for Evidentiary Integrity [24].

Multi-Modal Detection:

The next generation of detection will be more than pixel analysis - an Audio-Visual fusion approach [48, 58] and other multimedia-enabled [10] detection approaches to see discrepancies across streams.

Ethical and Privacy Issue:

Finally, this study is ethical in nature. The detection algorithms need access to user information which raises privacy concerns [45]. Ultimately, a future balance must be struck.

4.8 Cost-Benefit Analysis

4.8.1 Costs of AI and Deep Learning

The necessity to create and investigate AI/deep learning options suggests that facial forgery detection will need both technological and financial efforts. For instance, investigators had to recreate the Hybrid Optimized Deep Feature Fusion (HODFF-DD) model based on known data from personal and GPU efforts. It's reasonable to obtain smaller, geographically-investigative data for less time, but at some point, digging into large, widely-used datasets - FaceForensics++, Celeb-DF, FakeAVCeleb - takes a lot of GPU hours spread across networks. Compilation and work have to be set up hours in advance. Yes, creators should be transparent about their time taken between entry points - two hours for tests, four for analysis, etc. - but a collaborative effort is required.

Three costs are acknowledged which increase during the investigation's lifetime.

- **Processing power:** The need for a top of the line GPU system or cloud people/instances to process a deepfake detection model on video inclusion for training. The longer the project goes, the more energy output without consideration for maintenance/compounding schema-based systems that are on a schedule.
- **Data accessibility:** Large multimodal files need increasing access, redundancies and reviews for ethical private information. New multimodal entry points appear as time goes on.
- **Model viability:** New deepfake creators come out every day, and if the detector goes too long without change and update, it's retraining efforts (with larger datasets extensible) in the future - all of which are time, money, energy consuming.
- **Personnel:** There are always human system and implementation limitations out there that require trained experts. There's a cost to time as well as compounded coursework/learning that includes implementation to ensure it's worth the while.
- **Policy and ethics:** Even within a college setting, real/synthesized faces are needed to maintain a level of privacy and ethical ideals recognized through IRB

approval - which costs time through administration - and sometimes out-of-house assists.

The costs are substantial, but they're reasonable costs of development that lend to a replicable, reputable study and AI that can ultimately benefit the public.

4.8.2 Costs of Blockchain

Similar anticipated costs and logistical ease of sustainability occur when blockchain is the responsible detection that's integrated with AI. Although it's multimodal sources that can be tracked with authorship certainty for their quasi-integration, it needs development from initial programming to sustained decentralized access - often regulatory compliant, too.

Three major costs anticipated in three forms:

- **Creation/deployment:** Aspects of blockchain nodes and smart-contract logic needs interface from the AI detector to the blockchain ledger which requires specialized development and initial runtime systems.
- **Storage fees:** Every multimodal source is tracked and compounded comes networked access and storage while transaction, or gas fees accumulate - compounded on an emergent network over time.
- **Energy reliability:** The consensus mechanisms can be energy exhausting; the most secure option has a higher share of powered logistics that keep nodes powered.
- **Software adjustments:** Time and updates made on software, bugs found and securities of any blockchain endeavor requires hours from developers and extensive testing compounded over long stretches of time.
- **Interoperability:** Ensuring blockchain verifiability works with existing databases/APIs for social networking has outside costs down the line if they get more complicated down the road.

The anticipated costs are more technology based, but it's beneficial to assess these to find whether credibility increases are worth it or if integration isn't worth it due to sustainability.

4.8.3 Return On Investment (ROI)

Given that this is a university-researched event, ROI is less about financial gain and more about learned returns, technological findings and sociological implications. Overall, profitability is what's gained, established and usable down the line.

Returns include:

- **Educator benefits:** The team got real-world experience creating, compiling, and training data management and neuronal networks - not something available widely in other projects which further AI or cybersecurity inquiries later on for students/projects.
- **Scholarly contributions:** The recreation and verification of HODFF-DD is another thing for hybrid detection inquiry for deepfakes and a team-inoperable reference for others down the line.
- **Technological benefits:** This study shows that contributions from deep learning substantiated by other methods lends greater confidence for newly found conditioned media.
- **Sociological advantages:** Facial forgery detection protects people from identity theft purposes and privacy erosion; it's nice to know that public trust shouldn't be tarnished by mediative creations when something real should be real.
- **Institutional advancements:** This supports ASU's mission of engaged responsible AI developments while interdisciplinary efforts promote future grant opportunities and student engagement in progressive fields for collaboration.
- **Community access:** Findings/tools can lend to further developments that give people access to media viability before they can share/interact/use anything else - preventing nefariously created information/devices from entering the public eye.

It seems that anticipated income during this study is nonexistent but qualitative ROI. The time spent between processing, acquiring resources, and engaging talent brings potential inquiry down the line where evidence-based experiments boast feel-good social results. What can be learned will be worth its weight in gold in the future as AI statistics often reveal digital viability struggles.

4.9 User-experience and acceptance

4.9.1 User Feedback and Satisfaction

Not just technical precision but user trust, usability and perceived performance sustain any deepfake detection system's operation. For example, a study on user experience (UX) for AI-driven media integrity solutions found that transparency, interpretability and responsivity comprise effective user satisfaction determinants. For example, users want detectors that respond with justification based on *visual features* or *confidence scores* rather than simple “real/fake” disparity [1][10]. Consequently, trusted transparency in automation is even more critical for social media and professional environments where what can be seen isn't always trusted, and user-based investigations may not suffice.

According to [13][4], **responsiveness** is the highest relevant factor to user satisfaction. Users demand detection systems to detect manipulated videos as quickly as possible; systems designed for social media/in mobile environments should take only a few minutes at most. The faster a model can provide a determination, the better; thus deep learning architectures based on quick inference [16] and CNNs based on mobile implementation [34] create integrity without sacrifice. Furthermore, [6] found that users feel more supported and systems are more reliable when a user can see how the system found a deepfake (i.e., interpretable heat maps, overlays).

Testing detection systems in applied circumstances based on [9][11] found that successful detection with brief text results is preferred after detection systems are employed. If detection happens, levels of satisfaction increase when the user is informed why their media was called fake (i.e., different facial landmarks, lighting adjustment, eye movement misalignment). Systems based upon explainable AI gain this

interpretability through perceived reliability based on solutions without technical overtones which would otherwise promote such systems to science-reliant audiences. Instead, journalists, educators and policymakers would all benefit from such reliable systems when detectability is transparent and easily understood.

4.9.2 Addressing User Concerns and Expectations

Regardless of educated awareness built around deepfake deception, users cite concerns of **accuracy**, **privacy**, and **equity** - some of which inevitably hinder automated detection systems' acceptance. Surveys compiled by [10][2] confirm user concerns that false positive outcomes which approve authenticity would never exist, as would false negatives that allow manipulated fakes to permeate situations without feedback to users. This user dissatisfaction is compiled from sense-based positions which compel data-driven solutions to avoid absolute vulnerability. Thus, firm model validation and communication are critical to convey margin of error levels and recognition for continually trained datasets based on changeable deception methods.

Privacy-preserving detection systems are also recommended. According to [1][14], facial recognition patterns - biometric data - are a primary detection system component; thus, data should be handled **securely and in an anonymized manner**. Ideally, detection remains anonymous on the user end, or a blockchain audit system would ensure user identity is never connected or stored as hostile data following auditing systems for detection reasons. Such integrated systems are approved for good system design according to user expectations which reveal ethical AI should incorporate privacy concerns.

Furthermore, users expect detection systems to be **user-friendly, inclusive and unbiased**. [19][11] confirm that models trained on diverse biometric datasets are generally fairer and more reliable than those selected for demographic purposes in a training capacity. Without diverse datasets, all systems fail acceptance across any internationally-based system. Furthermore, multilingual integration, accessibility features and transparent open-sourcing further facilitate trust building.

Ultimately, systems that successfully justify truth relative to accuracy provided by internal human feedback systems are ideal. Systems that communicate certainty of detection, protect user privacy and ensure continued usability - performance through group protection and situational suggestion - comply with various criteria put forth in relative criteria review [1][6][10][14]. Explainable and responsive systems foster the necessary **socio-cultural integrity** to apply such tools in social media, law enforcement and general information settings.

4.10 Comparative Analysis with Blockchain-Based Solutions

The most recent developments in an architecture based on blockchain address the idea of digital media integrity where a tamper-proof distributed ledger involves storing the hashes of images and videos as well as model outputs and attribution details. While such architectures enhance registration and attribution values across the board, they represent a more retrospective, rather than real-time, approach to forgery detection.

However, our Hybrid Optimized Dual Feature Fusion Deepfake Detector (HODFF-DD) is proactive and relies upon deep-learning constructs that feature dual spatial feature extractors (InceptionResNet V1 - VGGFace2 - InceptionResNet V2 - ImageNet) with BiLSTM layers for temporal modeling, projecting an architecture that emphasizes intrinsic visual and temporal artifacts of manipulation for both frame- and video-level classification in real time.

Such systems based on blockchain (see Mastoi et al. [7], Colaco et al. [8]) ascertained cryptographic registration and subsequent registration efforts. They do not determine whether a manipulation is new or old, but rather, if the image or video is the same as the one authenticated and previously logged. In addition, the required registration process and authenticated determination requires a great amount of lag time and storage expenditures that are not feasible for scaled-down streaming efforts with inefficient, impractical facilitation.

The HODFF-DD model only takes into account the data at hand without any need for external efforts at correlation. In addition, the fusion of spatial and temporal factors

allows it to learn to generalize for a new generation of deepfake efforts with real-time classification based on actionable statistics: Accuracy, AUC, TPR, and TNR. The efficiency of the construction for implementation - as demonstrated through our API via Flask - for real-time inference with confidence overlays also adds inoperable efficacy.

While blockchain created environments are useful for ensuring honest material creation and reporting efforts, outside of a blockchain environment, detection is not possible. Yet detection is much more preferable with this state-of-the-art solution which provides a much greater ability and autonomy to flag potential manipulation for review before any such action would ever be taken on any chain. Thus, with deepfakes, HODFF-DD is a preemptive avenue of detection that instantly recognizes recitable content faster than would ever be possible with more predominantly blockchain-based systems.

4.11 Challenges Encountered

While implementing the development and implementation of this project, multiple challenges emerged relative to data and modeling implementation.

- **Data Imbalance and Generalization:** There is data imbalance between real and manipulated images across all benchmarking datasets from FaceForensics++, FakeAVCeleb and Celeb-DF. At first, due to data imbalance, the learning model was more inclined towards majority classes. WeightedRandomSampler, Rotation, Flip and Brightness augmentation were used to alleviate the situation.
- **Temporal Feature Learning:** Features between videos were hard to model due to too many facial expressions and lights over time and ultimately a network of BiLSTM(128→64) for the learnable segment was chosen out of trial and error across many recurrent networks.
- **Computational and Memory Limitations:** Models (Inception-ResNet V1 and V2) utilize a lot of GPU memory and take too long to train. The convolutional layers were frozen for the first epochs along with Spotted Hyena Optimization Technique to improve training convergence without impact on performance to mitigate these challenges.

- **Face Detection and Alignment Limitations:** It was hard to ensure that the thousands of videos processed all had cropped faces in the same area. The MTCNN detector struggled with extreme yawning/poses and occlusions. The solution was to implement a contextual 20 pixel margin and resize all frames to be set up at 300×300 pixels.
- **Evaluation & Deployment Limitations:** Post-processing was required for consistent results from video-level by averaging frame-level predictions. Majority voting allowed us to reduce this challenge while also supporting API development efforts post frame level prediction. Good latency for overlaying results on the video was needed for real-time efficiency.

Yet with a general goal of accuracy and responsiveness for varied datasets, all challenges were addressed.

5. Conclusions and Recommendations

5.1 Summary of Detailed Results

The Hybrid Optimized Dual Feature Fusion Deepfake Detector (HODFF-DD) was tested on various benchmarking datasets such as Celeb-DF v2, FaceForensics++ (FF++), as well as its manipulation sub-datasets (DeepFakes, FaceSwap, Face2Face, FaceShifter, and NeuralTextures). The model was designed to fuse InceptionResNet V1 (VGGFace2) and InceptionResNet V2 (ImageNet), with an added BiLSTM (128 → 64) temporal module and trained with the Spotted Hyena Optimizer with binary cross-

results on each dataset from the HODFF-DD pipeline (obtained by majority voting over frame-level results with evaluate.py):

Dataset	Accuracy	Precision	Recall	F1 Score
Celeb DF v2	94.5	90.29	93.07	91.66

DeepFakes(FF++)	93.3	89.86	91.6	90.73
FaceSwap(FF++)	93	89.34	91.34	90.33
Face2Face(FF++)	92.1	88.14	89.64	88.88
FaceShifter(FF++)	84.9	80.87	82.07	81.47
NeuralTextures(FF++)	78.65	74.22	68.14	71.08

Table 1: Performance of the HODFF-DD Model on Benchmark Datasets

Observations

- With an accuracy of 94.5%, our model's highest overall performance on Celeb-DF v2 showed resilience against realistic face swap forgeries.
- On important FF++ subsets like DeepFakes, FaceSwap, and Face2Face, the detection accuracy was consistently over 90%, demonstrating how effective this architecture is at producing images with multiple modifications.
- Due to high-frequency illumination errors that do not mirror those during facial warping processes, a slight degradation was seen on NeuralTextures.

Performance Characteristics:

The HODFF-DD model maintained an appropriate balance between spatial accuracy and temporal coherence caused by its dual-backbone fusion and BiLSTM encoding strategy. The HODFF-DD model further improved detection precision by ≈ 2.5-3% over the baseline models that use only the CNN architecture. The role of temporal modeling added to increased robustness over extended video data.

Deployment Results:

The final, quantized model was capable of real-time prediction with ≈ 23 FPS on RTX 3060 with the aid of a Flask-based REST API (serve.py), which allowed real-time overlaying of prediction labels and confidence levels on processed videos. In

conclusion, the results clearly show that HODFF-DD offers a detection solution that is generally more adaptive and practical to use compared to blockchain-based validation platforms.

5.2 Recommendations for facial forgery detection

Facial forgery detection research points to combining blockchain technology with AI for the most promising approach to ensure authenticity in digital media. Deep learning techniques further improved this capability on minute changes in lighting, texture patterns, and facial expressions using convolutional and attention-based networks. Sadhya and Qi (2023) illustrated that in recognizing faked face regions, models that focus on fine visual cues outperform prior CNN techniques [20]. In a related context, Kumar et al. (2024) demonstrated that in complex scenarios as well, it is possible for deepfake faces to be detected by ensemble machine learning models with reasonable accuracy [32]. However, Malik et al. (2022) mentioned that without secure verification techniques, these systems on theair own cannot guarantee long-term reliability as the production of synthetic media advances beyond forensics techniques [19].

5.2.1 Recommendations

1. **Adopt Hybrid AI–Blockchain Frameworks**
   Face forgery detection should rely on deep learning assured principles but application to the blockchain should possess assured accessibility. Mastoi et al. (2025) advocate for face detection during assurance and irreversibility on the blockchain while Patil et al. (2021) advocate for detection post-assurance through verification and global irreversibility [7], [43].

2. **Enhance Dataset Diversity and Realism**
   Many models flop when the forgery perpetrators do not emulate the same transformations they performed in training. Many works like Sadhya and Qi (2023) suggest that datasets should generalize and realize blurring, compression and low light elements [20]. The more generalized the datasets, the less bias

occurs at real life intersections.

3. **Develop Lightweight Verification Protocols**
   Gas fees, according to Colaco et al. (2025) and Verma et al. (2025) complicate development and scalability of transaction processing [8], [50]. Researchers should find consensus models that are light or hybrid which can be discovered under performance intensive standards to assure real time development seamlessly.

4. **Prioritize Privacy-Preserving Methods**
   Storing biometric data in any blockchain raises ethical/legal issues. Models will need to rely on encryption and zero-knowledge proofs moving forward which authenticate facial information but keep verification transparent.

5. **Integrate Explainable AI and Forensic Visualization**
   Systems should not only tell what's real or fake but how it got there. Forensic forgery detection needs visualizations like heat maps and saliency models so that professional investigators and end users alike find the forensic forgery detection interpretable.

6. **Encourage Cross-Platform Collaboration**
   According to Patil et al. (2021) and Costales et al. (2023) interoperability is important [43], [31]. Similar authenticity assessment standards should be distributed across social networks so if something is deemed manipulated or forged on one platform, it should not be able to enter another without such detection.

5.2.2 Future Research Directions in Facial Forgery Detection

Facial forgery detection is an emerging research trend with timely studies altering successful detection forges and interdisciplinary AI and blockchain verifications; however, with deepfakes always on the move, time sensitive research must delve into

hybridized, usable, ethical systems beyond single performance detection models to apply in real world manipulations.

1. **Integration of Multimodal Detection**

Integrated systems need to assess multiple forensic cues - not just visual like auditory, behavioral - and Malik et al. (2022) note that thus far reliable systems come from facial features or texture features; cross-modal deepfakes involving video AND speech reduce detection due to blending [19]. Visual forensics can work with auditory forensics for a stronger pipeline.

2. **Privacy-Preserving Verification Models**

The biometric element will most likely impact future blockchain involvement; Colaco et al. (2025) call for a privacy-preserving blockchain ecosystem which authenticates records but not biometric data storage [8]. Future studies will find homomorphic encryption and zero knowledge applications to transparent systems with no prying eyes.

3. **Decentralized Federated Learning**

Systems need deepfake learning everywhere which suggests face acquisition is not necessary; however, Verma et al. (2025) note the possibility for federated learning where the data stays with the user but the learning model only learns from their data and shares variables with the blockchain community [50]. This supports collaborative consent without compromising ownership.

4. **Real-Time Detection for Streaming Platforms**

As video attribution becomes real-time generated, systems must create support in real time; Costales et al. (2023) champion an embedded model relying on blockchain smart contracts for real-time video streaming authenticity assessment [31]. Further studies will make this palpable from lightweight algorithms giving real-time notifications without expansive time delays.

5. **Ethical Governance and Standardization**

Beyond non-technical ethics of governance, future studies will extend beyond standards championed by Mastoi et al. (2025) for non-operational concerns [7]. Standardized solutions backed by blockchain verification will inform all at a social media/law network level so standardized ethics to assess inclusion of facial data, log justified rationale, modeling accountability will all be imperative to prevent bias for public trust/inclusive fairness.

5.2.3 Summary

Facially forged media will ultimately be sensed from a range of algorithmic awareness to human post verified digital reality. Detection will rely upon deep learning but ultimately be able to lean upon blockchain to maintain consistent histories and identity protections over time. Thus, the ultimate binary detection will align with technological effectiveness and humane, ethics-based, design-based sensibilities of the future to make sure that what people see, they can ultimately trust again - and rely upon a level of accountability for digital realities of what's considered true in their minds' eyes at all times.

5.3 Future Prospects and Ongoing Research

Synthetic-media forensics is a moving field and with developments in deep learning and blockchain-based provenance, the following are how platforms attempt to safeguard authenticity better. Surveys and timely systems papers note three trends: (i) reliable, transferable detectors that are subject to generator updates and compression transfers; (ii) adversarially resilient, explainable pipelines that lend credibility to users and courts; (iii) Audit-verifiable provenance via content fingerprints and detector attestations positioned on tamper-evident ledgers. [1][2][4][6][7][8][10][11][14][19][23][31][50]

5.3.1 Emerging Trends in Information Security

As impersonation generated by AI proliferates, these trends extend the secure surface area from **model resiliency** to **system level assurance**:

- **Adversarial-Resilient Detection.** The exploration of adversarial training, ensemble defenses, feature-space regularization motivated by studies on 2D-Malafide show that popular face-deepfake detectors are vulnerable; vulnerability studies request investigation. [17][11][10]
- **Explainable & Auditable Forensics.** Where moderation is denied, the appeal and scrutiny of a court render saliency/attention maps, decision fusion and confidence calibration imperative. [11][10][1]
- **Privacy-Preserving Assurance.** Emphasis on on-device inference, hashed/hashed IDs, and ledgers that can be audited but not contain the raw media to secure biometric verification privacy. [1][14][7][31]
- **Blockchain-Backed Provenance.** Smart-contract interfaces employ cryptographic hashes and detector decisions across nodes/detectors (DDS) for collective-intelligence approaches utilizing permissioned+public systems with low-cost high integrity/success solutions or high transparency systems. [7][8][23][26][31][50]
- **Standards & Interoperability.** The potential for a universal authenticity schema is debated, along with cross-compatible attestations through platforms to secure forensic assessments on re-posting and reformatting. [23][31][43][50]

5.3.2 Advancements in AI and Blockchain for Social Platforms

Where **spatiotemporal modeling**, **frequency cues** and **ledger anchoring** converge:

- **Hybrid Video Models.** 3D attention networks and CNN-LSTM/Transformer pipelines typically yield better results than single stage baselines for spatio-temporal rendering. [6][27][29][30][16]
- **Frequency-Aware & Attention-Guided Cues.** The combination of attention and FFT/frequency band detection makes these more robust to compression and low-quality uploads frequently found in social feeds. [18][20][15]
- **Specialized Temporal/Behavioral Signals.** With optical flow artifacts and eye movement. These are precursors to on-the-go detection or live/streamed scenarios. [12][13]

- **Multimodal & Ensemble Learning.** Audio-visual coupling with dynamic weighting and ensemble (i.e., weighted average) makes these more robust to unseen creators. [48][47]
- **Edge/Real-Time Inference.** Architecture suited for low latency detection in production settings and live calls/streams. [4][5][16]
- **Ledger-Anchored Verification.** Result (hashes, labels, probabilities) and the subsequent moderation/appeal process recorded in smart contracts for non-repudiation and chain-of-custody purposes. [7][8][23][24][31][50]

5.3.3 Continuing Research and Development

The following areas warrant continued R&D:

- **Generalizability to New Generators.** Report cross-dataset accuracy and domain-shift statistics; create **zero-/few-shot** detectors that flag unseen manipulations. [10][19][42]
- **Data-Centric Resilience.** Create harder benchmarks like codec stress, low light, motion blur, and social-platform transformations; evaluate resistance to post-processing. [10][19]
- **Privacy-Preserving & Federated Learning.** Work with **hybrid blockchains** that hash and attest to data instead of biometric data. Create models where the data resides (creators, platforms) and only share model updates. [50][26][7][31]
- **Provenance Upon Upload.** Monitor integrity and tampering rates with validation & latency and cost; hash and attest upon upload to give every piece of media its own identifiable and verifiable stamp of authenticity. [7][23][31][50]
- **Human-Centered, Explainable Forensics.** Confidence reporting, saliency explanations, and reviewer workflows should be standard to increase user trust and reduce the negative impact of false positive reporting. [1][10][11]
- **Policy, Ethics & Governance.** Detected and logged provenance should exist with fair-use, consent, and data protection requirements. Effective interoperable schemas must be invented for cross platform authenticity attributions. [7][14][31][43]

The union of multimodal, transformer era detectors and **blockchain-based provenance** will be the baseline across the field. Systems operating in conjunction offer the most reliable deepfake prevention at scale across platforms through adversarially robust modeling, privacy-preserving authentication and interoperable authenticity attribution. [6][7][10][17][23][31]

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